

# RAJESH KANADE

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## PROFESSIONAL SUMMARY

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Generative AI Engineer with 1+ year of experience in designing, developing, and deploying production-grade LLM applications, RAG pipelines, and agentic AI systems. Skilled in LangChain, LangGraph, OpenAI, Anthropic, Gemini APIs, FastAPI, and cloud-native microservices. Experienced in prompt engineering, structured outputs using Pydantic and JSON Schema, semantic search, vector databases including FAISS, ChromaDB, and Pinecone, and scalable AI integration for enterprise SaaS platforms. Proficient in building async APIs, AI-powered automation workflows, and enterprise-grade backend systems with strong knowledge of LLM observability, evaluation, CI/CD pipelines, and Agile/Scrum development environments, backed by co-authored BERT-based NLP research.

## WORK EXPERIENCE

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### Generative AI Engineer / Full Stack Developer

Studium Tech Pvt. Ltd. | Pune

June 2025 – Present

- Designed and deployed production-grade LLM pipelines using LangChain, RAG, and OpenAI/Gemini APIs integrated into enterprise SaaS workflows serving thousands of users.
- Built agentic AI workflows with LangChain tool-calling and agent loops for multi-step reasoning, dynamic document search, and knowledge retrieval — aligned to AutoGen/CrewAI patterns.
- Implemented RAG architectures (chunking → embedding → vector store → retrieval) with ChromaDB, achieving sub-2s query response time and ~40% improvement in answer relevance.
- Engineered prompt versioning and structured output strategies (Pydantic, JSON schema) to reduce LLM hallucinations by ~30% and standardize downstream parsing.
- Built scalable async FastAPI and Node.js microservices for AI/LLM API integration into multi-tenant SaaS systems with JWT-based auth and RBAC.
- Integrated LangSmith for LLM observability, token/cost tracking, and LLM pipeline debugging.
- Collaborated in Agile/Scrum teams with CI/CD pipelines (GitHub Actions), Git version control, and cross-functional delivery alongside Solution Architects and domain developers.

## TECHNICAL SKILLS

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- **GenAI / LLMs:** LangChain, LangGraph, OpenAI API, Anthropic API, Gemini API, Azure OpenAI, Hugging Face, Ollama, Prompt Engineering, Prompt Versioning, Structured Output (Pydantic, JSON Schema), Function Calling, Tool Use, Agentic Workflows
- **RAG & Vector Databases:** RAG Pipeline Design, Chunking, Embeddings, Semantic Search, BM25 Hybrid Search, FAISS, ChromaDB, Pinecone, pgvector, Weaviate
- **Agentic Frameworks:** LangChain Agents, LangGraph, AutoGen, CrewAI, MCP Server/Client, Coding Agents, Review Agents
- **Evaluation & Guardrails:** Ragas, LangSmith, DeepEval, Guardrails AI, NeMo Guardrails, Custom Validators, LangFuse, Token & Cost Observability
- **Backend & APIs:** Python, FastAPI (Async), Node.js, Fastify, REST APIs, gRPC, Microservices, JWT Authentication, RBAC
- **Cloud & DevOps:** AWS S3, Azure Services, GCP Services, Git, GitHub Actions, GitOps/FluxCD, CI/CD, PM2
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis (Caching, Pub/Sub)
- **Frontend:** React.js, Next.js, TypeScript, Material UI, Tailwind CSS, Streamlit
- **Domain Knowledge:** Domain-Driven Design (DDD), Multi-tenant SaaS, Enterprise Integrations (Jira, Confluence, GitHub, Artifactory)

## GENERATIVE AI PROJECTS

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### AI Web Page Assistant & Video Explainer — Chrome Extension

*Tech Stack: Python, LangChain, LangGraph, OpenAI API, FastAPI, ChromaDB, React.js, Pydantic*

- Designed end-to-end RAG pipeline with ChromaDB for webpage/video transcript ingestion, improving answer relevance ~40% vs. direct LLM calls.
- Applied prompt versioning and Pydantic-enforced structured output, reducing hallucinations ~30% and enabling reliable downstream parsing.
- Developed 10+ async FastAPI endpoints for content ingestion, embedding generation, and query handling with sub-2s response time.
- Integrated OpenAI API for real-time summarization, contextual Q&A, and multi-modal content analysis (text + video transcripts).
- Added LangSmith tracing for observability — token usage dashboards and latency profiling across agent steps.

### Multi-Modal Document Intelligence Platform

*Tech Stack: Python, LangChain, AutoGen, OpenAI API, FastAPI, ChromaDB, Redis, Streamlit*

- Implemented hybrid search (semantic + BM25) over ChromaDB with Redis caching, reducing query latency ~45% at enterprise scale.
- Built agentic reasoning loop (AutoGen-style) for multi-document cross-referencing and intelligent summarization.
- Designed async FastAPI APIs for concurrent document uploads, embedding pipelines, and enterprise-scale query throughput.
- Implemented RBAC-secured document access with Guardrails AI validators for output safety and Streamlit UI for real-time querying.
- Integrated DeepEval for automated RAG quality evaluation — precision, recall, and faithfulness scoring per query.

### Accreditation Insight Generator

*Tech Stack: Python, LangChain, LangGraph, OpenAI API, FastAPI, PostgreSQL, Pydantic*

- Built LLM-driven gap analysis system over structured JSON data using LangGraph state machines, improving accreditation reporting efficiency ~70%.
- Designed structured output schema (Pydantic) ensuring consistent JSON responses for downstream React UI consumption.
- Developed coding-agent-style workflow for automated evidence mapping and report generation, cutting manual analysis time ~60%.
- Integrated enterprise data sources (PostgreSQL, JSON config files) via custom MCP-style tool connectors in LangChain.

## EDUCATION

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### Master of Computer Applications (MCA)

Fergusson College, Pune University | CGPA: 9.10

2024 – 2026

### Bachelor of Science (Computer Science)

Savitribai Phule Pune University | CGPA: 8.87

2021 – 2024

## CERTIFICATIONS & LANGUAGES

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**Certifications:** React JS — Apna College | JavaScript (Intermediate) — HackerRank | Python Developer — Udemy

**Languages:** English (Professional) | Hindi (Native) | Marathi (Native)